Data Structures and Algorithms 2

Question 5: Swarm Algorithms

GitHub repository: https://github.com/janadatatech/dsa2-coursework

Coursework Question 5 (10 marks)
Read the two papers presented in Unit 6.2:

a) What are the possible behaviours modelled in each paper? Explain each distinct behaviour in a couple of lines.

b) Both articles compare their models with other, pre-existing models. Choose one pre-existing model they have in common and compare both models' results against it.

Q5(a) Behaviours modelled in each paper

The Cat Swarm Optimization is explained with each cat representing a possible solution. The algorithm uses two behaviours called seeking and tracing. The seeking mode is represented as a cat resting while remaining alert and the algorithm makes small changes to copies of its position, then checking their fitness scores and selecting a position. The tracing mode is represented by a cat chasing a target and the algorithm updates the cat's speed and position towards the best solution found at that time (Chu, Tsai and Pan, 2006).

The Grey Wolf Optimizer is where each wolf represents a possible solution. The algorithm models leadership and hunting behaviours which is split into the three best solutions called alpha, beta and delta. They guide the remaining wolves, which are called omega. The wolves spread out to explore different areas to search for prey and this helps the algorithm find other possible solutions. The wolves change their positions depending on what the location of the prey is estimated to be, in the algorithm, this means searching around solutions that look promising. The wolves use information from the alpha, beta and delta to choose their next positions and the movements become smaller as the search progresses. This helps the wolves focus on improving the solutions that they're homing in on (Mirjalili, Mirjalili and Lewis, 2014).

Q5(b): Comparison against Particle Swarm Optimization

Both papers compare their algorithm with the Particle Swarm Optimization so for comparison I use this as the common algorithm and compare how well each method finds solutions in the reported tests. Chu, Tsai and Pan (2006) compared the Cat Swarm Optimization with two versions of Particle Swarm Optimization using six test functions and they reported that the Cat Swarm found better solutions than both Particle Swarm versions. However, they only show a results graph for one test so this limits how much of the reported improvement that can be checked. Mirjalili, Mirjalili and Lewis (2014) ran the Gray Wolf Optimizer 30 times on each test function, and their results show that it performed better than Particle Swarm on some tests, but the Particle Swarm also performing better on others.

Test function	Average score for Gray Wolf Optimization	Average score for Particle Swarm Optimization
F5	26.81258	96.71832
F6	0.816579	0.000102

These tests aim to find the lowest score and the results show that the Gray Wolf Optimization gives the better average result on F5, while Particle Swarm gives the better result on F6. It shows that performance depends on the problem being tested (Mirjalili, Mirjalili and Lewis, 2014).

Overall, The authors of the Cat Swarm Optimization report better results than the Particle Swarm Optimization across their six tests. The Gray Wolf performs better than Particle Swarm on some tests, but Particle Swarm performs better on others, which shows that an algorithm's performance depends on the problem. This means you cannot directly compare Cat Swarm Optimization with Gray Wolf Optimization because the papers state that different test settings are used.

References

Chu, S.-C., Tsai, P.-W. and Pan, J.-S. (2006) 'Cat swarm optimization', in Yang, Q. and Webb, G. (eds.) PRICAI 2006: Trends in Artificial Intelligence. Berlin, Heidelberg: Springer, pp. 854–858. Available at: https://doi.org/10.1007/11801603_94.

Mirjalili, S., Mirjalili, S.M. and Lewis, A. (2014) 'Grey Wolf Optimizer', Advances in Engineering Software, 69, pp. 46–61. Available at: https://doi.org/10.1016/j.advengsoft.2013.12.007.